

Presentation 2

- Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallCategorization_Oct23_Group2.pdf

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20integrity%20checks/CallDurationComparison_22Oct_HaileeKim.R

(For data import:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_R_CarolineCorr.R)

- What did you do?

I coded and created a Median Call Duration by Hour graph to visualize and validate the consistency of inbound call durations between the CAR and All Calls datasets. This process involved cleaning and filtering both datasets to include only records after March 15, 2025, standardizing timestamps to the America/Chicago timezone, and calculating duration metrics for each dataset. I summarized the median call duration per hour, combined the two sources, and plotted them together for comparison. In addition to the coding work, I also prepared presentation slides for Objective 1.2, clearly illustrating the overlap between datasets and explaining the rationale behind key data processing choices.

- How does it help the project?

This work directly supports Objective 1, which aims to ensure consistency and reliability in inbound call duration reporting across multiple data sources. By comparing CAR and All Calls datasets after March 15, 2025, I confirmed that both show aligned hourly patterns, suggesting strong data integrity and synchronization between systems. The analysis revealed that median call durations peak around 8 AM - corresponding to the start of operational hours - and gradually decline toward the end of the day, which reflects expected operational trends. Additionally, I verified that differences between `answer_time` and `start_time` in the All Calls dataset have minimal impact, simplifying future analysis of call timing variables. Finally, I identified that refining the hour variable logic in the CAR dataset - using start hour before 17:00 and end hour afterward - improves classification accuracy for extended evening calls.

- Issues faced (if any)

N/A

- Attempts to resolve issues (if any)

N/A

- Issues resolved (if any)

N/A

- Next steps

Durations during operational hours showed strong alignment between the two datasets, while some discrepancies appeared outside of these hours. To improve overall accuracy, the next step is to further investigate calls occurring during non-operational periods to understand how they are being captured, timestamped, and classified. This deeper review will help identify whether these differences stem from system logging behaviors, after-hours routing processes, or data extraction inconsistencies.

- References (Mention if you built up on someone else's work)

Presentation 1

- Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Data%20integrity%20checks/CARAllCallsIntegrity_7Oct_EDATeam2.pdf

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20integrity%20checks/AllInboundCalls_6Oct_CarolineCorr.R

- What did you do?

I pulled relevant code from the previous quarter and developed strategies to tackle Objective 1 (comparing and visualizing the number of calls between the CAR dataset and the corresponding inbound calls in the All Calls dataset). Then I worked with team members to import the files correctly and code the first plot. I also designed the team presentation and compiled the emerging questions that highlighted the issues we faced and informed potential next steps. After the Friday check-in meeting, I added additional comments on handling daylight savings using the `with_tz` function, and on creating a `month_year` column to facilitate joining and comparing tables.

- How does it help the project?

It led to a solid visualization of Objective 1 and helped identify potential reasons for the remaining mismatches between the All Calls and CAR datasets. Also, it contributed to making our work more reproducible.

- Issues faced (if any)

Our team initially struggled to correctly read the data into R. We divided into two groups based on our preferred programming languages - I was on the R team with Caroline and Nikole. We followed the approach we had learned in our sequence class, where all datasets are stored in the same local folder and imported as needed. For this reason, we pushed the datasets to GitHub and worked from the `/data` repository.

After our Friday check-in meeting, we learned that it was possible to incorporate the OneDrive location in the Jupyter notebooks. However, we encountered the error message "The specified directory does not exist," and we are still troubleshooting this recommended but unfamiliar setup. Because the combined dataset was too large to include in the repository and we wanted to avoid the confusion of multiple people editing the same R scripts simultaneously - a problem

we had faced in our sequence class - we decided to code collaboratively on one laptop (Caroline's). All of us met in person at Main and worked together throughout the process.

- Attempts to resolve issues (if any)

I pulled relevant code from the previous quarter of how to import the datasets in R and we also consulted ChatGPT to help edit the code to include CSVs. We also decided to download the All Calls data locally from Vivienne's version in a shared Google drive instead of starting that R import code from scratch.

- Issues resolved (if any)

We were able to import the data but we were told from Krish that we must not push your datasets to GitHub, so fundamentally we're still figuring out how to correctly import the datasets.

- Next steps

After the pre-grading feedback, we've filtered 6 relevant numbers to CAR data. To address the remaining mismatch in inbound call numbers, we plan to further explore whether understanding the category names and filtering by "User Type" could improve accuracy. We can also take a closer look at the timestamp discrepancies between "Activity Start Timestamp" and "Start Time/Answer Time."

- References (Mention if you built up on someone else's work)

We used the R data import code from spring quarter to get started loading in the CAR data.

Link:

https://github.com/arvindkrishna87/STAT390_SP25_LegalAid/blob/main/Presentations/explore.R